Unit 2 Quiz C

Forces & Translational Dynamics · AP-style multiple choice

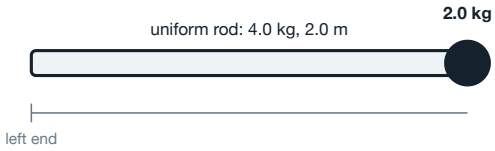
12 questions + 1 optional Time: **24 min** (AP pace ≈ 2 min each) Calculator OK g = 9.8 m/s²

- Before choosing, **write one line of work** in the margin: an FBD sketch, an equation, or a ratio. Those are the habits the FRQs grade.
- Circle your confidence for every question. A lucky guess still counts as a gap.
- Grade with the coaching guide. For every miss AND every 'guess', read the full walkthrough and do the 'Go deeper' question.

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1. `2.1 Center of mass`

A uniform 2.0 m rod of mass 4.0 kg has a 2.0 kg ball fixed to its right end. How far from the LEFT end is the system's center of mass?



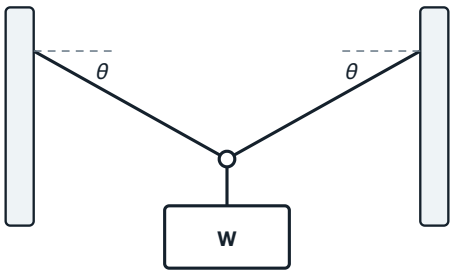
A 1.0 m **B** 1.33 m **C** 1.5 m **D** 1.67 m

Confidence: sure | 50/50 | guess Answer: ____

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2. `2.4 Equilibrium (2-D)`

A sign of weight W hangs from two cables. Each cable makes 30° with the horizontal. What is the tension in EACH cable?



A W/2 **B** 0.58 W **C** W **D** 2W

Confidence: sure | 50/50 | guess Answer: ____

3. `2.3 Newton's 3rd law`

On ice skates, you push on a wall and glide away from it. Which force accelerates you?

A The force you exert on the wall

B The force the wall exerts on you

C The internal force of your muscles

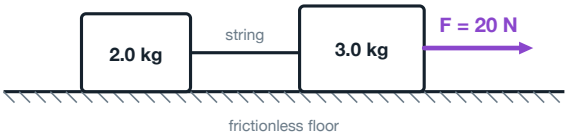
D No force; you glide because of inertia

Confidence: sure | 50/50 | guess Answer: ____

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4. `2.5 Connected systems`

A 2.0 kg block and a 3.0 kg block on a frictionless floor are tied together with a light string. A 20 N force pulls the 3.0 kg block. What is the tension in the string?



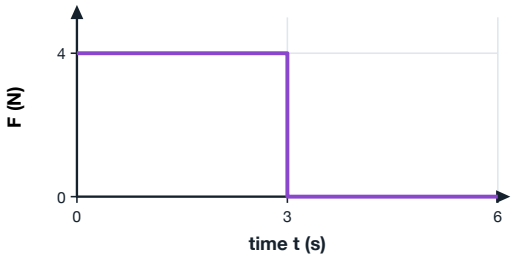
A 4.0 N **B** 8.0 N **C** 12 N **D** 20 N

Confidence: sure | 50/50 | guess Answer: ____

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5. `2.5 Force → motion graphs`

A cart starts at rest on a frictionless track. The graph shows the net force on it. Which describes the cart's velocity?



A Increases linearly from 0 to 3 s, then stays constant

B Constant from 0 to 3 s, then drops to zero

C Increases from 0 to 3 s, then decreases back to zero

D Zero until 3 s, then increases

Confidence: sure | 50/50 | guess Answer: ____

6. 2.6 Mass vs weight

A 70 kg astronaut stands on the Moon (g_{Moon} = 1.6 N/kg). What are her mass and weight there?

A 70 kg; 686 N

B 11.4 kg; 112 N

C 70 kg; 112 N

D 70 kg; 0 N

Confidence: sure 50/50 guess Answer: ____

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7. 2.6 Free fall

An astronaut on the Moon (no air) drops a hammer and a feather from the same height at the same time. They land together. The best explanation:

A There's no gravity on the Moon without air

B The gravitational force on each is the same

C The gravitational force is proportional to mass, so a = F/m is the same for both

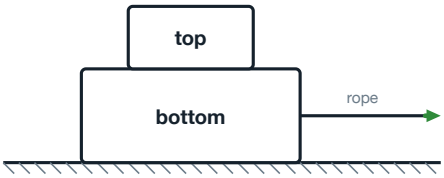
D The hammer's larger inertia exactly cancels the feather's smaller weight

Confidence: sure 50/50 guess Answer: ____

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8. 2.7 Friction (stacked)

A small block rests on top of a larger block. A rope pulls the bottom block so that BOTH move together at CONSTANT velocity. What is the friction force on the top block?



A Zero

B Forward, equal to $\mu_s m_{top} g$

C Backward, equal to $\mu_k m_{top} g$

D Forward, equal to the rope tension

Confidence: sure 50/50 guess Answer: ____

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9. 2.7 Friction + angles

You move a sled across rough snow with the same force F at the same angle θ, either PULLING (rope angled up) or PUSHING (handle angled down). Which gives the larger acceleration?

A Pulling, because the normal force (and so friction) is smaller

B Pushing, because the normal force is larger, giving more grip

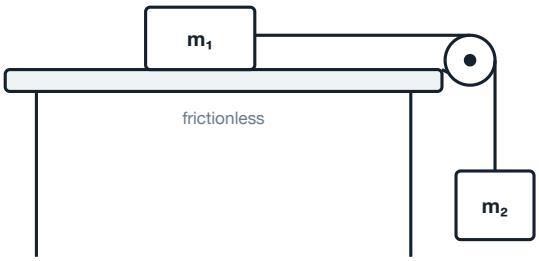
C Both the same, since F and θ are the same

D Pushing, because pushing transfers more force

Confidence: sure 50/50 guess Answer: ____

10. 2.5 Modified Atwood

Block m_1 sits on a frictionless table. A string runs over an ideal pulley at the edge to block m_2, which hangs. What is the acceleration of the blocks?

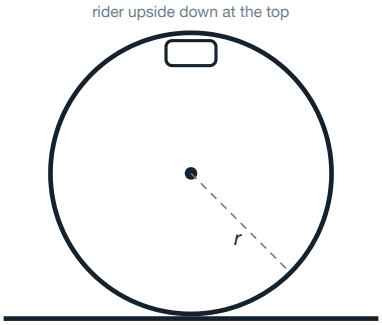


A m_2g / m_1 **B** $m_2g / (m_1 + m_2)$ **C** $(m_2 - m_1)g / (m_1 + m_2)$ **D** g

Confidence: sure 50/50 guess Answer: ____

11. 2.9 Vertical loop

A 50 kg rider goes through the top of a loop of radius 10 m at 14 m/s, upside down. What is the normal force from the seat on the rider?



A 490 N, downward **B** 1470 N, downward **C** 490 N, upward **D** 0 N

Confidence: sure 50/50 guess Answer: ____

12. 2.9 + 2.7 Rotor ride · BOSS

In a 'rotor' ride, riders stand against the wall of a spinning cylinder of radius 2.5 m, and the floor drops away. $\mu_s = 0.40$ between the rider and the wall. What is the minimum wall speed that keeps riders from sliding down?

A 3.1 m/s **B** 4.9 m/s **C** 7.8 m/s **D** 61 m/s

Confidence: sure 50/50 guess Answer: ____

13. `2.8 Springs · optional: ask if springs are on the test`

A spring stretches 0.20 m when a 2.0 kg mass hangs from it on Earth. The same spring and mass are taken to the Moon (g = 1.6 N/kg). The stretch on the Moon is:

A 0.20 m, since the mass is the same

B About 0.033 m

C About 1.2 m

D Zero

Confidence: sure 50/50 guess Answer: ____